Two toxins sharing the C-C-CC-C-C cysteine framework: cys2 reads them as the same string, cys_hp3 tells them apart and still shows where the cysteines are, and hp_pbotc_1st_ed2 tells them apart but hides the cysteines among the other hydrophobic residues.
Conotoxin MiXIVA (Conus miles, snail, UniProt P06751) and U1-theraphotoxin-Pm1a (Pterinochilus murinus, spider, P0DQZ2), 36 residues each, from 5124 mature peptides in ConoServer and Swiss-Prot.

🟧 position holding a cysteine in both peptides
— the two peptides encode to the same symbol here

a

cys2   c = C, x = the 19 others

residue number	1									10										20										30							
snail residues	E	C	R	E	K	G	Q	G	C	T	N	T	A	L	C	C	P	G	L	E	C	E	G	Q	S	Q	G	G	C	A	W	T	I	S	G	V	
snail encoded	x	c	x	x	x	x	x	x	c	x	x	x	x	x	c	c	x	x	x	x	c	x	x	x	x	x	x	x	c	x	x	x	x	x	x	x	
36/36 same symbol	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	|	22/22 windows identical at k=15
spider encoded	x	c	x	x	x	x	x	x	c	x	x	x	x	x	c	c	x	x	x	x	c	x	x	x	x	x	x	x	c	x	x	x	x	x	x	x	
spider residues	E	C	R	W	F	W	G	G	C	N	N	D	A	D	C	C	K	H	L	E	C	K	R	K	W	P	H	I	C	L	W	D	G	T	F	T	

Every position agrees, so cys2 cannot tell these two peptides apart at all.

b

hp_pbotc_1st_ed_hpc3 (cys_hp3)   h = AFILMPVWY, p = DEGHKNQRST, c = C

snail residues	E	C	R	E	K	G	Q	G	C	T	N	T	A	L	C	C	P	G	L	E	C	E	G	Q	S	Q	G	G	C	A	W	T	I	S	G	V	
snail encoded	p	c	p	p	p	p	p	p	c	p	p	p	h	h	c	c	h	p	h	p	c	p	p	p	p	p	p	p	c	h	h	p	h	p	p	h	
25/36 same symbol	|	|	|				|	|	|	|	|	|	|		|	|		|	|	|	|	|	|	|			|		|	|	|	|		|			0/22 windows identical at k=15
spider encoded	p	c	p	h	h	h	p	p	c	p	p	p	h	p	c	c	p	p	h	p	c	p	p	p	h	h	p	h	c	h	h	p	p	p	h	p	
spider residues	E	C	R	W	F	W	G	G	C	N	N	D	A	D	C	C	K	H	L	E	C	K	R	K	W	P	H	I	C	L	W	D	G	T	F	T	

The six c symbols are the framework; the h and p between them tell the peptides apart.

c

hp_pbotc_1st_ed2   h = ACFILMPVWY (C among them), p = DEGHKNQRST

snail residues	E	C	R	E	K	G	Q	G	C	T	N	T	A	L	C	C	P	G	L	E	C	E	G	Q	S	Q	G	G	C	A	W	T	I	S	G	V	
snail encoded	p	h	p	p	p	p	p	p	h	p	p	p	h	h	h	h	h	p	h	p	h	p	p	p	p	p	p	p	h	h	h	p	h	p	p	h	
25/36 same symbol	|	|	|				|	|	|	|	|	|	|		|	|		|	|	|	|	|	|	|			|		|	|	|	|		|			0/22 windows identical at k=15
spider encoded	p	h	p	h	h	h	p	p	h	p	p	p	h	p	h	h	p	p	h	p	h	p	p	p	h	h	p	h	h	h	h	p	p	p	h	p	
spider residues	E	C	R	W	F	W	G	G	C	N	N	D	A	D	C	C	K	H	L	E	C	K	R	K	W	P	H	I	C	L	W	D	G	T	F	T	

Tells them apart just as well, but the cysteines are 6 of 14 h in the snail and 6 of 17 in the spider, so the framework cannot be read off the string.